

Institute of Science and Technology (IST)
B.Sc (Hons.) in CSE Part-1, 2nd Semester Preparation Test
CSE-510223: Discrete Mathematics
Time Duration: 3 hours Full Marks: 80

[N.B. -- The figures in the right margin indicates full marks. All parts including subsections must be answered sequentially]

Answer any four of the following

1. a) What is disjunction and conjunction? Discuss with truth table and example. 4
- b) How can this English sentence be translated into a logical expression? 4
 - i) "You can access the Internet from campus only if you are a computer science major or you are not a freshman.
 - ii) "Every student in this class has studied calculus" using predicates and quantifiers
- c) Differentiate i) universal and existential quantifiers, ii) tautology and contradiction. 4
- d) Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent. 4
- e) Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences. 4

2. a) Define cardinality of a set, infinite set and power set. What is the power set of the empty set and of the set $\{2,3,4\}$? 5
- b) If a and r are real numbers and $r \neq 0$, then prove that 5

$$\sum_{j=0}^n ar^j = \frac{ar^{n+1}-a}{r-1} \text{ when } r \neq 1$$

$$(n+1)a \text{ when } r = 1$$
- c) Define one to one and onto functions. 3
- d) Let f be a function from $\{a, b, c, d\}$ to $\{10, 15, 20, 25\}$ with $f(a)=20, f(b)=15, f(c)=25$ and $f(d)=10$. Is f a bijection? 3
- e) Give a direct proof that if n is an integer and $3n+2$ is odd, then n is odd. 4

3. a) How many different bit string of length eight either start with a 1 or end with the two bits 00? 4
- b) Let $f(x) = a^n x^n + a^{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_0, a_1, \dots, a_{n-1}, a_n$ are real numbers. Then $f(x)$ is $O(x^n)$. 5
- c) Define relation with example. Write down the properties of relation? 6
- d) What are closures of relations? Let relation $R = \{(1,3), (1,4), (2,1), (3,2)\}$ on the set $A = \{1,2,3,4\}$ is not transitive. What is transitive closure of relation R ? 5

4. a) What is graph? What are the types of graph, brief with example? 5
- b) Draw the precedence graph for the following expressions:- 2

$S_1 \ a:=0$
 $S_2 \ a:=a+1$
 $S_3 \ b:=a+2$
 $S_4 \ c:=b+a$
 $S_5 \ d:=c+1$

- c) Prove that an undirected graph has an even number of vertices of odd degree. 4
- d) How can a graph be represented? Give example. 6
- e) Draw i) $K_{3,5}$ ii) Q_3 and iii) W_5 ? 3
5. a) Define m-ary and full m-ary tree with example. 4
- b) i) What is the postfix form of the expression $((x + y) \uparrow 2) + ((x - 4)/3)$? 6
 ii) What is the value of the prefix expression $+ * 235 / \uparrow 2 3 4$?
- c) Prove that there are at most m^h leaves in an m-ary tree of length h. 5
- d) What is spanning cost tree. Using Prim's algorithm find the minimum cost spanning route for the following graph. 1+4=5

	2	b	3	c	1	d
a						
3		1		2		5
e	4	f	3	g	3	h
4		2		4		3
	3		3		1	
i		j		k		m

6. a) Construct half adder with truth table. 5
- b) Using Karnaugh maps to simplify the sum of products expressions 4
 (a) $xyz + x\bar{y}z + \bar{x}yz + \bar{x}\bar{y}z$.
- c) Construct circuit for the following expression using logic gates. 6
 i) $xz + \bar{x}yz$ ii) $(x+z) + \bar{x}$
- d) Find the values of the Boolean function represented by $F(x, y, z) = xy + z$. 5